

Aftab Hossain Mikat

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About

Senior Computer Science and Engineering undergraduate at Rajshahi University of Engineering and Technology (RUET), working at the intersection of Multimodal AI and Natural Language Processing. My research focuses on document-grounded visual reasoning, vision-language models for Bangla and other low-resource languages, and efficient pretraining of compact language models. I enjoy connecting research ideas with reproducible experiments and practical software systems.

Education

Rajshahi University of Engineering and Technology (RUET) *Dec 2022 – Feb 2027*
B.Sc. in Computer Science and Engineering (Expected) *Rajshahi, Bangladesh*

- **CGPA:** 3.39/4.00.
- **Focus:** Natural Language Processing, Small Language Models, Deep Learning, Algorithms, and Software Engineering.

St. Joseph Higher Secondary School *Jul 2019 – Jun 2022*
Higher Secondary *Dhaka, Bangladesh*

- **GPA:** 5.00/5.00.

Publications

When the Evidence Is Right, the Math Is Easy: Understanding the Challenges of Document-Grounded Reasoning *Sep 2026*

Aftab Hossain Mikat, Tamjid Hasan Fahim

Accepted archival paper and poster at the DocInsights Workshop @ EMNLP 2026 – DocSem Shared Task
[OpenReview](#) 

- Developed LILIPUT, a zero-shot pipeline that separates visual evidence localization from numerical reasoning; achieved 49.54% joint exact accuracy and 66.76% macro evidence F1 on 1,730 held-out test instances.
- Analysis identified document understanding and evidence extraction, rather than final arithmetic, as the primary bottleneck in document-grounded quantitative reasoning.

Dense Supervision, Gentle Decay: Curriculum-Guided Rebalancing of ELECTRA-Style RTD and MLM for Small-Data Pretraining *Aug 2026*

Aftab Hossain Mikat, Tamjid Hasan Fahim

Accepted at the BabyLM Workshop @ EMNLP 2026 – Strict-Small Track

[Preprint](#) 

- Investigated curriculum-guided objective rebalancing between ELECTRA-style Replaced Token Detection (RTD) and Masked Language Modeling (MLM) for small-data pretraining.
- Focused on compute-aware experimentation and efficient supervision for compact language models.

Experience

Walton Hi-Tech Industries PLC *Feb 2026 – Mar 2026*
ICT Department - Two-week Industrial Attachment *Gazipur, Bangladesh*

- Gained practical exposure to enterprise software, server infrastructure, networking, manufacturing technology, and quality-control operations across industrial departments.
- Collaborated in a team to build a Notification Management Website using React, TypeScript, Vite, and Django; the system received positive feedback from the organization.

Projects

BoostLearning AI: AI-Powered Study Assistant

[GitHub](#) 

- Developed a full-stack study assistant featuring a customizable lab-report generator, PDF/text-to-MCQ and narrative question generation, topic explanation, paraphrasing, and personalized study planning.
- **Tech Stack:** React, TypeScript, Flask, Firebase Authentication, MongoDB, Gemini API

Notification Management Website

[GitHub](#) 

- Built with a team during the Walton industrial attachment to support structured creation and management of organizational notifications through a modern web interface.
- **Tech Stack:** React, TypeScript, Vite, Django

WhatsApp Chat Analyzer: Communication Analytics Platform

[GitHub](#) 

- Developed a web-based communication analytics platform with NLP/ML-powered sentiment analysis, message classification, multilingual support, interactive visualizations, and secure JWT/Bcrypt authentication; optimized large-chat processing through caching and batch translation.
- **Tech Stack:** Python, Streamlit, NLP/ML, JWT, Bcrypt

MovieZone: Movie Discovery Platform

[Live](#)  – [GitHub](#) 

- Developed a React and TypeScript movie discovery platform with search, genre and country filtering, ratings-based sorting, detailed movie information, personalized watchlists, recommendations, and dark mode. Integrated an external API for rich film and series data.
- **Tech Stack:** React, TypeScript, External Movie API

Full-Stack Blog Platform

[Live](#)  – [GitHub](#) 

- Developed a full-stack blogging platform for publishing and reading long-form content with a responsive user experience and persistent data storage.
- **Tech Stack:** Node.js, React, TypeScript, MongoDB

Technical Skills

Programming	C++, Python, JavaScript, TypeScript
AI and ML	PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, NLP, Computer Vision, Multimodal AI, Small Language Models, Deep Learning
Web	React, Vite, Node.js, Express.js, Flask, Django, HTML, CSS, Bootstrap
Databases	MongoDB, MySQL, Firebase
Tools	Git, GitHub, LaTeX, Kaggle, Vercel, Railway
CS Foundations	Data Structures and Algorithms, Competitive Programming, Algorithmic Problem Solving

Research Interests and Achievements

Research Interests: Multimodal AI, Document Understanding, Visual Evidence Localization, Vision-Language Learning, Bangla and Low-resource VLMs, Small Language Models, Efficient Language Model Pretraining, Transformers, MLM, ELECTRA/RTD, and Curriculum Learning.

Achievements: Two papers accepted at EMNLP 2026 workshops: DocInsights (DocSem Shared Task, archival poster) and BabyLM (Strict-Small Track).

Additional Achievements

- Solved 250+ programming problems on Codeforces.
- Received General Scholarships in the Secondary School Certificate and Junior School Certificate examinations.
- **Languages:** Bengali and English.